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Market Design as Organizational Problem: Explaining System Failures in Platform Markets[1]

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System failures occur when organizations managing sociotechnical systems lose control over interactions between components. To prevent such failures, organizations decompose systems into simpler parts and empower frontline experts to manage them in a flat hierarchy. The article examines whether such "modularization" applies to the design of platform markets, focusing on California's electricity markets between 1995 and 2000. Although modularization seemed like a reasonable technique to avoid system failures, the platforms became vulnerable to persistent gaming. Drawing on rich archival data and interviews, the article finds that the platform markets violated a central requirement for distributed organizations. Market actors, unlike organizational employees, did not cooperate to maintain the relationship between modules. This hindered designers' ability to prevent, detect, and correct failures. The case reveals a tension between designers' efforts to coordinate actors via incentives and organizational strategies to manage complex sociotechnical systems, highlighting the need for an organizational sociology of digital marketplaces.

INTRODUCTION

Platform markets increasingly mediate economic transactions (Stark and Pais 2020). They organize retail trading, match cabdrivers to passengers,

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and assign ad space to clients. Beyond the mundane, they help with the allocation of public goods such as food donations, public school slots, and emission certificates. Despite the rhetoric of free markets, these digital marketplaces are not domains of unfettered competition. Instead, they are complex sociotechnical systems (Hughes 1987) where technical and social processes interact to solve a problem. Platforms have architectures (Lei 2021)—the rules, procedures, and interfaces of the platform software—that carefully orchestrate the interactions between market participants. These architectures are built and maintained by technical experts, who work inside organizations (Aspers, Bengtsson, and Dobeson 2020; Rilinger 2024*a*; Stark and Broeck 2024).

Some of these experts, so-called market designers (Breslau 2011; Viljoen, Goldenfein, and McGuigan 2021; Rilinger 2024*a*), pursue a project of social engineering. They draw on tools of algorithmic management (Kellogg, Valentine, and Christin 2020) and software engineering (Burrell and Fourcade 2020) to nudge market actors toward desired behavior (Yeung 2017; Curchod et al. 2020; Rahman, Weiss, and Karunakaran 2023). When they succeed, the market actors follow the logic of a theoretical market mechanism, and the platform solves a specific allocation problem. It finds the cheapest assignment of taxi cabs to users, the most equitable distribution of school children to school slots, or the least wasteful allocation of water rights (Kominers, Teytelboym, and Crawford 2017; Rilinger 2024*b*). In short, platforms are sociotechnical systems the designers build to coordinate market actors in the hope of solving allocation problems.

However, designers also fail. Particularly markets that are supposed to help with the allocation of public goods like electricity, emission certificates, and water often give rise to behavior that derail designers' objectives. If this behavior is persistent and severe, the market may no longer provide the required solution to the underlying allocation problem (Narassimhan et al. 2018; Wheeler and Garrick 2020; Cramton 2021). If the problematic behavior responds to incentives and opportunities in the rules, procedures, and interfaces of the platform, such a situation qualifies as a *system failure*. System failures are by-products of how a complex sociotechnical system is organized. They emerge when the organization is no longer able to understand the complex interactions between system elements in time to intervene should something go wrong (Perrow [1984] 1999; Leveson 2011; Clearfield

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and Tilcsik 2018; Vaughan 2021, 2023). When the platform architecture gives rise to incentives and opportunities for activities that derail designers' goals, they thus qualify as a system failure.

Existing sociological research has long asked why such failures emerge in complex sociotechnical systems (Pinch 1991; Vaughan 1999) and seems to offer useful intuitions about how they might be avoided. A key insight is that technical experts at the "coalface" of the organization (Huising and Silbey 2018; Cantu et al. 2020) are best able to react to the complex and surprising interactions between system elements. To avoid system failures, organizations need to simultaneously empower and simplify the work of these experts. Multiple literatures suggest that this can be achieved by breaking the sociotechnical system into loosely connected "modules" that can be built by flexible teams in flat hierarchies (Simon 1962; Sanchez and Mahoney 1996; Perrow 1999; Baldwin and Clark 2000). Conversely, classic bureaucracies with long divisions of labor and centralized authority tend to remove discretion from technical experts. They are therefore less well suited to build and manage complex sociotechnical systems that need to adapt to uncertain and volatile environments quickly (Vaughan [1996] 2016).

Yet few studies have explored whether these insights carry over to the project of designing digital markets. Recent work has mainly studied how politics influences market design processes (Rilinger 2022; MacKenzie, Caliskan, and Rommerskirchen 2023) and how transaction platforms shape the relations among participants (Shestakofsky and Kelkar 2020; Lei 2021; Rahman 2021; Rahman et al. 2023). Following calls to investigate those who build digital infrastructures (Bailey and Barley 2020; Christin 2020; Rilinger 2023*a*, 2024*a*), this article turns to the organizational challenges of platform design. Specifically, I study an extreme case where market designers did what existing theory would suggest. That is, to make a complex design problem tractable, they built and managed a system of interconnected submarkets in a modular division of labor. Yet they produced a platform architecture that was prone to persistent forms of gaming.

To explain this puzzle, the article argues that there is a tension between the mode of coordination in platform markets and the organizational requirements for modularization. Modularization requires that *employees* cooperate to actively maintain preestablished boundaries between the different parts of the system. *In contrast to employees*, market actors have no particular reason to collaborate with designers. Market designers coordinate market actors via incentives; they create rules, procedures, and interfaces that make it beneficial for market actors to act in line with designers' objectives. If it is in their interest, market actors may actively undermine the intended relationship between different parts of the system. Moreover, they look for profitable trades in any part of the system. This creates problems for modular design work. The case will demonstrate how design decisions in

supposedly unrelated parts of the system can now develop problematic consequences for each other and how the modular division of labor complicates market designers' ability to detect and fix these consequences. In short, a basic strategy to make the complexity of the market system tractable to designers—decomposing the system into simpler parts—can produce the very system failures it is meant to avoid.

I develop this theoretical argument through an empirical analysis of system failures in auctions markets at the heart of California's first electricity markets. Technical experts—economists, utility officials, software engineers, and system operations researchers—built the digital infrastructure for these transaction systems between 1995 and 1998 in an organization called Western Power Exchange (WEPEX). Throughout their existence, the auctions were prone to a variety of manipulative strategies that derailed designers' objectives. I identify the design flaws that produced the incentives and opportunities for these strategies and trace them to decisions during the design process. I then explore why it was difficult for designers to resolve these issues after they discovered the games during the operation of the market. At each turn, I show how these issues can be traced back to a basic tension between the logic of modularization and the mode of coordination in platform markets.

A final section draws on secondary literature about other cases to consider the scope conditions of this theoretical insight. California's energy crisis was an extreme case where gaming was particularly detrimental to designers' goals. Accordingly, the case does not reveal an overriding cause of system failures. Other platform markets may be less vulnerable to gaming or offset problems of modularization. Yet the extreme case strips aways these mitigating factors and reveals what may otherwise be less apparent: Although modularization seems like an obvious choice to make the complexity of platform markets tractable to designers, it does not actually work as advertised in this context. It is more likely to cause rather than to prevent system failures. This insight has implications not just for research on system failures and modularization but for organizational and economic sociology more generally. It suggests the need for an economic sociology that treats digital marketplaces as organizational forms and explores how their unique mode of coordination relates to the distributional outcomes they produce.

System Failures and Modularization in Other Domains

The concept of system failures originates in the literature on organizational accidents and how to avoid them (Pinch 1991; Vaughan 1999). This literature has largely focused on organizations that manage safety-critical technologies like nuclear power plants, dams, and military technology (Vaughan 1999; Leveson 2011; Le Coze 2015). In contrast to earlier work, which

identified the overriding causes of accidents, this literature has long taken a structural point of view. It examines how the organization of sociotechnical systems itself can lead to accidents (Hughes 1987). Charles Perrow famously suggested that technical systems that display high "interactive complexity" and "tight coupling" will eventually produce interactions among component failures that cascade out of control before anyone can intervene. Interactive complexity means that the interactions between system elements are not linear. Instead, the system may change its behavior and produce surprising sequences of events. "Tight coupling" means that there is not much time to intervene should something go wrong. In such systems, catastrophic accidents may be rare but are ultimately unavoidable, or normal (Perrow [1984] 1999; Clearfield and Tilcsik 2018).

Although the hypothesis of normal accidents has long been contested (Leveson et al. 2009), Perrow's argument has drawn attention to the epistemic nature of system failures (Downer 2011). Whether or not a system is overly complex does not simply depend on the structure of the technology itself. It also depends on the monitoring capabilities of the organization managing it. System failures emerge, therefore, when *those in charge of the technology* become unable to comprehend problematic interactions in time to intervene before something goes wrong. A sociological explanation of system failures therefore needs to focus on the interplay between the technology and the larger "system that includes history, competition, scarcity, bureaucratic procedures, power, rules and norms, hierarchy, culture, and patterns of information" (Vaughan [1996] 2016, p. 415).

Through countless case studies, the literature has developed a complex and nuanced vocabulary to describe the organizational mechanisms behind system failures. A basic lesson is that problems emerge when organizational structures remove authority over critical decisions from frontline operators and technical experts who are closest to risky technologies (Weick, Sutcliffe, and Obstfeld 1999; Grabowski and Roberts 2019; Cantu et al. 2020). It is air traffic controllers (La Porte 1996; Vaughan 2021), crews on flight decks (Rochlin, La Porte, and Roberts 1998), operators in nuclear power plants (Roberts 1990), and similar experts who need to react to unanticipated events quickly and appropriately (Weick et al. 1999, p. 36). In long divisions of labor with centralized decision-making, it simply takes too long for signals of trouble to percolate to the top. Meanwhile, interpretative processes can distort these signals as they move from one level of the hierarchy to the next (Farjoun and Starbuck 2007). Similarly, errors spread more easily in tightly integrated bureaucracies with strict superior-subordinate relations than in loosely coupled structures where teams pursue different strategies for their work (Weick 1998; Pidgeon and O'Leary 2000). Accordingly, high-reliability organizations deploy fluid organizational structures where flexible teams of experts can quickly react to the complex interactions of system elements (La Porte 1996).

However, to empower technical experts and frontline operators, it is important that their work does not cognitively overwhelm them (Perrow 1999). A very common strategy to achieve this goal is called "modularization" (Baldwin and Clark 2000; Benassi 2009; Brusoni et al. 2023). Safety engineers decompose technological systems into interconnected subsystems or "modules" that are related to each other via interfaces. An interface is an abstraction that selects only those elements of a module that are relevant to the others. Each module performs a function for the larger system, and the interfaces specify how the modules are related to each other. Within a system of interface relations, the modules then become exchangeable. One implementation can be swapped out for another if the interface relations are preserved. Even more importantly, modules are buffered from each other (Perrow 1999). Interactions between elements are complex within the module but limited and standardized between modules (Baldwin and Clark 2000).

The resulting system architecture can now be implemented and managed in a division of labor that "mirrors" this architecture (Colfer and Baldwin 2016). Flexible teams can take charge of individual modules, coordinating along the lines of the interface relationships between modules. As long as teams working on module A know what inputs they receive and what outputs they have to produce for module B, they can ignore how module B works. This has advantages for the speed and reliability of the design process (Baldwin and Clark 2000, p. 64). Each team can focus on the comparatively simple task of creating a module that fits the parameters of the larger system. The organization can adopt flat hierarchies, because the impact of teams' work on the rest of the system is clearly circumscribed. Accordingly, relatively little centralized coordination is necessary. With short pathways for decision-making, teams can quickly react to surprises and adjust their approach flexibly (Lee and Edmondson 2017).

Air traffic control offers a compelling illustration of this logic. Controllers manage a complex and dynamic system of airplanes that navigate in the sky. Because this complexity would overwhelm a single operator, the designers of control towers divide the airspace into areas and each area into sectors. Standard operating protocols establish what information needs to be passed on as airplanes move from one sector to the next. Relatively independent workstations can now manage individual sectors, allowing operators to focus on only a subset of the system (Vaughan 2021, p. 196f.). Standardized protocols for interactions between workstations enable rapid and silent communication between those who manage adjacent sectors—that is, clear input and output relationships between modules enable coordination between teams whose attention is focused on their individual sectors (Vaughan 2021, p.198). The modular structure of the airspace enables a modular division of labor that makes the complexity of air traffic tractable to controllers.

We might expect this logic to apply to market design as well. Indeed, most platform markets for complex allocation problems employ modular architectures (Baldwin and Woodard 2009; Jacobides, Cennamo, and Gawer 2018; Cusumano, Gawer, and Yoffie 2019; Gawer 2021; Kretschmer et al. 2022). They consist of multiple submarkets and technical systems that are loosely interconnected with each other and jointly produce a desired allocation. For instance, Upwork is a platform that matches freelancers and clients for specific projects. In one part of the platform architecture, freelancers search for jobs posted by clients. In another part, clients search for freelancer profiles. Together, the two markets (one for jobs, one for freelancers) identify the optimal allocation of skilled freelancers to clients. The dynamics in one market therefore need to inform matching in the other.

The fundamental problem behind system failures in platform markets is also similar to technical systems: Market designers create systems of rules, procedures, and interfaces that guide actors toward a desired form of calculative behavior (Wilson 1998; Yeung 2017). If the interplay between these elements becomes overly complex, designers may be unable to comprehend the incentives that actors face in different situations (Kominers and Teytelboym 2020). Unintended interactions between rules, procedures, and interfaces can therefore produce surprising behavior that violates designers' goals and may be difficult to detect either ex ante or ex post. Research in the sociology of markets has therefore applied the theory of normal accidents in technological systems to certain markets—high interactive complexity and tight coupling of market behavior overwhelm regulatory efforts to manage the market (Lounsbury and Hirsch 2010; Palmer and Maher 2010; Clearfield and Tilcsik 2018; Min and Borch 2022).

Given these parallels between the design of technical systems and markets, we might expect the basic insight of research on modularization to carry over. Figure 1 illustrates such an organizational setup. The organization would build and manage its systems of digital markets by assembling flexible teams of designers in a flat hierarchy (bottom) that carefully build and manage individual market modules (middle) while coordinating along the preestablished interface relationships between these markets. The architecture of the platform market structures the division of labor and lines of coordination between teams. Indeed, the setup in figure 1 is quite common in platform companies. The principle of modular design is foundational to software development, and software companies—including platform companies—often adopt organizational paradigms like "Agile" or "Scrum" that emphasize flexible teamwork with wide discretion (Bahrami 2009; Highsmith 2009). This means that the organizations typically come to mirror the modular architecture of the software system in the structure of teamwork.

But even though such modularization seems like a reasonable choice to prevent system failures in platform markets and even though tech startups

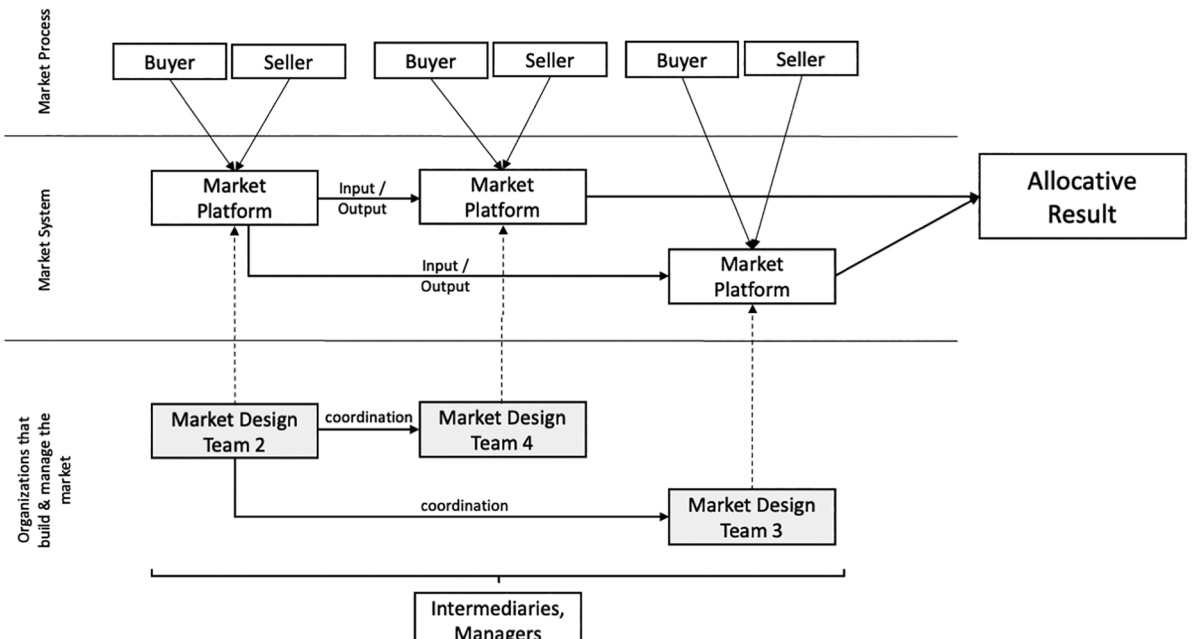


FIG. 1.—Illustration of modularization in platform organization

commonly choose this strategy, there are flaws in the underlying analogy between platform markets and other sociotechnical systems. In particular, researchers have not yet considered how the coordination in platform markets differs from that of managing other complex sociotechnical systems. As Perrow pointed out, accidents in technical systems go back to random interactions between component failures (Downer 2011). But market actors are not unthinking, mechanical parts of the system. Instead, they are reflexive actors who are trying to maximize profits (Perrow 2010). If the system of incentives makes behavior rational that can bring down the system, this behavior will occur not randomly, but quickly, secretly, and intentionally.

Market designers also deploy a distinct mode of coordination. Most tools to coordinate employees within regular organizations are not available to them. Regular organizations are hierarchies that are structured by employees' formal rights to make decisions and voice opinions on behalf of the firm (Coase 1988; Freeland and Zuckerman Sivan 2018). This authority structure provides managers with tools to organize the work, ranging from direct task assignment, supervision, and reporting routines (March and Simon 1958) to community-building tools that foster identification with the company's goals (Gibbons and Henderson 2012; Turco 2016). All these tools enable modes of coordination that are based on a *cooperative* relationship between the employees and the organization.

In contrast, market designers rely on a more limited and indirect mode of coordination. They structure actors' digital choice environments to nudge them toward the desired behavior. The rules, procedures, and interfaces of the transaction platform create a situation where actors find it *beneficial* to follow the required logic of action. They only do what is in their immediate interest (Wilson 1998; Roth 2018). The California case reveals how these differences matter for strategies to avoid system failures. Modularization depends on the stability of design rules. Before the creation of a new system, designers need to establish what the interface relationships between the different modules are. This allows teams to ignore those parts of the system that are not relevant to the work on their module. In organizations that manage complex technologies, this is possible because there are no special reasons to violate the modular division of labor and employees actively collaborate to maintain the boundaries between the different modules (Vaughan 2021).

However, in platform design, the relationship between the different pieces of the platform architecture—the different modules—partly depends on the trading activities of market participants. If it is in their interest, they will derail the intended relationship between the different modules. Because they will look for profitable trades in any part of the system, it is difficult for market designers to specify these relations ex ante or stabilize them cooperatively ex post. As the analysis will reveal, standard problems of technical design

work can therefore generate inconsistencies that create incentives and opportunities for behavior that derails the market mechanism.

Specifically, I will show that market actors' antagonism produced three problems for designers in California: Coordination between design teams remained incomplete (incomplete coordination), the oversight suffered from systematic blind spots (structural secrecy), and potential interactions between rules, procedures, and interfaces could always change (temporal instability). This undermined a foundational tool that organizations use to make complex technical systems tractable to those who need to manage and build them. As the last section will show, this finding poses general questions about the practical limits of market design and the organizational theories we can employ to understand problems in the administration of platform markets.

CASE, DATA, AND METHODS

This article examines an early case of electricity market design: the creation of California's first electricity markets between 1993 and 2000. Specifically, I focus on the design of several auction platforms by WEPEX and the organizations that succeeded it. The creation of these auctions was part of a larger effort to restructure California's electricity system around competitive markets. As is well known, the new system all but collapsed during the Western Energy Crisis of 2000–2001. Supply shortages drove up the prices, and sellers—most prominently the Texan Energy company Enron—manipulated the markets to extract riskless profits as the electricity system teetered on the brink of collapse (Weare 2003; Walsh 2004). While the energy crisis epitomizes the flaws in California's market system, it was also the result of many interdependent processes. Market dynamics, environmental factors, the development of supply and demand in the West, and political dynamics at the state and federal levels interacted to produce the crisis (Joskow 2001; Taylor et al. 2015; Rilinger 2024*b*).

To preserve the analytical focus on the organizational dimension of market design, I focus on a smaller part of the larger story: the design of digital auction markets between 1995 and 1998 as well as the period preceding the crisis between 1998 and 2000, when the auction markets experienced pervasive forms of destructive behavior. Figure 2 illustrates that the article focuses on technical design work and the period preceding the crisis.

This focus has three analytical advantages. First, we can study the organizational dynamics behind not just the creation but also the subsequent efforts to repair the market design. Second, the auction markets suffered from a fairly common problem: "illegal arbitrage games." Arbitrage trades exploit price differences between markets by simultaneously buying and selling in both markets. In other platform markets, such trades are forbidden

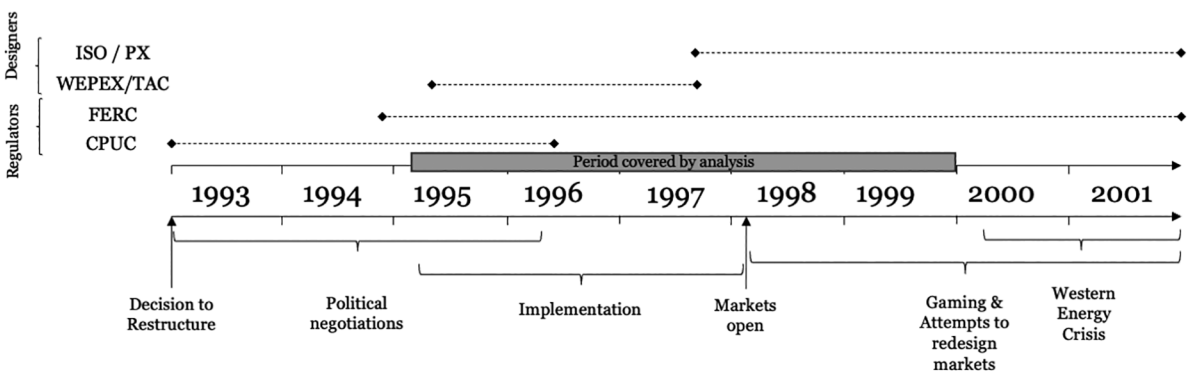


FIG. 2.—Timeline for design of California's first electricity markets

when the price differences do not go back to "market fundamentals"—information that is relevant to the allocation problem at hand—but to administrative differences between markets. Illegal arbitrage games circumvent these prohibitions and thereby derail the intended logic of action in both markets. In market design, this is a common problem because such games go back to inconsistencies in the administrative rules that govern the market process.[2] Third, the focus on an extreme case has certain analytical advantages. Not only were the markets very vulnerable to gaming, but there were also very few external protections against such gaming in place. The burden was thus very much on the market designers in WEPEX to make the platform design work within a thin margin of error. The case thus strips away mitigating factors and clearly displays the organizational challenges of modular platform design.[3]

The study relies on material from four different archives and a variety of supplementary materials that include 35 interviews with design experts. The first set of material comes from the archives of the Federal Energy Regulatory Commission (FERC), which assumed regulatory oversight of the new system in 1998 and accompanied the design process starting in 1996. I scraped the most important dockets for both market design and litigation. This yielded more than 17,500 submissions, ranging in length from two to several hundred pages. Table 1 summarizes the material from the FERC dockets.

Second, I relied on 15 congressional investigations that probed the California energy crisis. Staff of the congressional committees collected and evaluated evidence of market manipulation in California. Since FERC's actions also became subject to inquiry, the congressional investigations help offset some of the potential biases of the FERC archives.

The third set of data comes from two archives in California. The technical work of building the new markets started in August 1995 and lasted until March 1998. After the markets opened, the process of adjusting rules and software continued throughout the first two years of operation and into the crisis. Regulatory processes before the California Public Utility Commission (CPUC) accompanied each step of this process. Although jurisdiction shifted to FERC after the markets opened, the initial transformation of

[2] Controlling such metareflexive trading is a core problem in market design (Roth and Wilson 2019) and the foundation for definitions of market manipulation in financial markets (Taylor et al. 2015; Partnoy 2018). For energy markets, FERC justifies the prohibition of such trades like this: "Participants in energy markets should have as their price signal . . . supply and demand fundamentals, not ancillary considerations that bear no relation to underlying market fundamentals." FERC, "Staff White Paper on Anti-Market Manipulation Enforcement Efforts Ten Years After EPAct 2005," https://www.ferc.gov/sites/default/files/2020-05/marketmanipulationwhitepaper.pdf, accessed October 28, 2024.

[3] For a complete analysis of the California energy crisis, the other dimensions of market design failure, and related problems of regulatory capture, see Rilinger (2023*b*, 2024*b*).

TABLE 1
FERC Dockets, Litigation for California Energy Crisis

Main Subject	Docket Number	Description
Ancillary market crisis	ER98-2843	Crisis in ancillary markets at CAISO, consolidated docket. Contains all filings relevant to crisis and redesign of the markets.
Gaming case	EL03-180	Case that investigates the various manipulative games first discovered in the Enron memos
Enron investigation	PA02-2	Contains all material relevant to the investigation of manipulation in Western energy markets through Enron and others. In particular, evidence of transactions from all sellers in the Western energy markets.
Refund case	EL00-95 EL00-98	Investigation of market power abuses in the California energy markets between 2000 and 2001. Contains all settlements regarding market power abuses as well as the evolution of the oversight standards during the crises. Cross-references to all individual cases involving the Western energy crisis.
California parties' application for establishment of ISO/PX	ER96-1663 EC96-19	All material related to the design of the markets. In particular, WEPEX hearings that debate the correct approach to measuring market power and overseeing the California markets (1996–99) Contains all formal filings pertaining to the design and implementation of the California markets.
Development of orders concerning deregulation (592, 888, 2000)	RM96-6 RM95-8 RM99-2	Landmark decisions to open the transmission system in order to prepare the ground for deregulation (Order 888 in 1996) and the subsequent development of regional transmission organizations (Order 2000 in 1999). Contains transcripts that debate market power and methodology in energy markets.
Ancillary market crisis	ER98-2843	Crisis in ancillary markets at CAISO, consolidated docket. Contains all filings relevant to crisis and redesign of the markets.

the industry was shepherded by the CPUC. The archives of the CPUC in San Francisco therefore contain two vast dockets that chronicle the entire restructuring process until the crisis. These document internal debates, problems, and their solutions. During visits to this archive, I digitized 7,604 pages of filings; 4,134 of these were transcripts of public hearings, during which different design decisions were debated.

In 1997, the implementation work migrated into the California Independent System Operator (CAISO) and Power Exchange (PX), which had just been constituted as new organizations. In the aftermath of the crisis, the California State Archive in Sacramento collected business documents from different departments of both CAISO and the PX, dating from 1997 to 2001. Some are documents that the governing boards used to oversee the

implementation process. Others are reports and internal filings from the different departments that were responsible for the implementation of different elements of the design. During two visits in 2018 and 2019, I digitized 12,307 pages of documents. Tables 2 and 3 summarize the information from the different archives.

I supplemented the analysis with material from less systematic collections. Most importantly, the internet ArchiveX (https://archive.org) contains snapshots of the coordination website for the implementation process between 1995 and 1997. Although this material is quite limited (less than 100 pages), there are minutes from meetings of the steering committee that organized all technical work. This information was invaluable to reconstruct the organizational structure of the design process. In addition, I use secondary literature about the California energy crisis, gray literature on market design proposals by academics at Stanford and the Electricity Institute in Palo Alto, and the trade press, particularly *Utility Fortnightly*, and *The Electricity Journal*. To establish the descriptive statistics about price differences between the different markets, I used a quantitative dataset from the University of California Energy Institute containing prices and trading volumes in California's markets between 1998 and 2001.

To provide some context for the archival documents, I also conducted 35 in-depth interviews with market designers who worked on the implementation of the markets in California. These interviews are purely supplementary. With 20 years of distance, the risk of rationalization and hindsight bias is substantial, and I used none of the transcripts to establish facts about the design process. Instead, I used the interviews to clarify the meaning of archival documents and provide context. I followed the rules of historical case

TABLE 2
Archival Material (CPUC)

Archival Material	Description
CPUC order instituting rulemaking and order instituting investigation:	
CPUC Archives San Francisco, R.94-04-031 / I.94-04-132, boxes 10–50	All filings responding to the Blue Book that formally started the process of restructuring in 1994. Contains information on the political processes before the CPUC and reports about/filings from to technical WEPEX processes of implementation.
CPUC Archives San Francisco, R.94-04-031/I.94-04-132, boxes 50–53	CPUC's attempts to cope with the energy crisis (consolidated docket)
CPUC Archives San Francisco, R.94-04-031/I.94-04-132, boxes 69–71	Transcripts from hearings during the design process (1994–97).

TABLE 3
Archival Material (California State Archives)

	Archival Material	Description
California Independent System Operator (CAISO):		
Market surveillance committee, reports (1998–2001)	California State Archive, R400.007, boxes 12–14	Material (academic papers, analyses, data charts, etc.) related to the problems in the ancillary markets
Electricity Oversight Board, subject files (related to market power analysis at ISO, 1997–2001)	California State Archive, R400.005, boxes 4 and 5 R400.010, boxes 18 and 19	Background material pertaining to specific issues, (including market power in ancillary markets)
ISO Market Surveillance Committee and Board of Governor meeting files (1998–2000)	California State Archive, R400.006, box 6, folder 1, and box 9, folder 6	All materials used during board meetings, including all material presented to the subcommittees. Contains memos, memoranda, and status reports on problems with ancillary markets
Power Exchange (PX):		
Teaching material (market primer, etc.) for Board of Governors	Two reports (available online)	Background information on how the interface of the market works
Electricity Oversight Board, subject files . . .	California State Archive, R400.005, box 5, folders 10–25, R400.010, box 18, folder 1, and box 19, folder 14	Background material related to market power analysis at PX, 1997–2001
Board of Governors meeting files (1998–2000) . . .	California State Archive, R400.008, box 13, folder 15, and box 14, folder 9	Agenda, minutes, and background materials presented during meetings of the stakeholder board, including material of committee meetings, here, compliance
PX compliance filings . . .	George Sladoje's, PX CEO, personal archive	I received a DVD that contains detailed documents chronicling the PX's investigation of Enron's early gaming activities.

studies in analyzing these data (Abbott 2014; Yin 2017). See the methodological appendix for more details on the analytical approach.

ARGUMENT AND FINDINGS

The analysis proceeds in four parts. The first two sections develop the empirical puzzle at the heart of this article. I show that modularization seemed like a reasonable choice for the designers but produced the very problems it was meant to avoid. The third part develops the theoretical explanation in

conversation with the case. It traces why designers adopted problematic design decisions in the modular division of labor and suggests that these were near impossible to avoid, identify, or fix given that the designers were indirectly coordinating profit-seeking market actors rather than directly coordinating employees. This suggests an inherent tension between the coordination in platform markets and a basic technique to make complex systems tractable to designers. The last section goes beyond the California case to consider the scope conditions of this theoretical argument.

THE EMPIRICAL PUZZLE

Modularization as Reasonable Choice to Preempt System Failures

For the better part of a century, three large utilities produced, transmitted, and sold most of California's electricity to end users. In 1993, the governor's office and the CPUC decided to restructure this system around competitive markets. The utilities would have to divest their power plants and buy energy from power sellers in wholesale markets for electricity. A neutral entity would manage the transmission of electricity. At the time, the governor, Pete Wilson, had ambitions to run for president and imposed a tight deadline for the project. The new system was to come online by January 1, 1998 (Sweeney 2002).

Market designers had to figure out how to translate relatively broad political ideas into material reality. This work took place in an organization called "WEPEX." What started as a loose set of working groups in 1995 commissioned by the CPUC quickly turned into a formal organization under the leadership of the utilities. Because the utilities owned the transmission system and knew how to operate it, they dominated the executive level of WEPEX—the so-called steering committee.[4] In addition to the utility officials, WEPEX hired consultants, system operations experts, and academic market designers from Harvard, Berkeley, Arizona, CalTech, and MIT to help with the specification of the new system.

The designers faced a tremendous challenge. Because electricity flows at the speed of light on a vast transmission system and even minor imbalances can lead to blackouts, a so-called system operator has to closely manage the system in real time. Markets prepare the work of the system operator by identifying the optimal combination of generation resources to meet demand in a given hour in the future. Grid management, in turn, has to adjust and implement the market results on the basis of updated system information. Smooth operation necessarily requires constant, mutual adjustments between short-term markets and grid management. Typically, a single organization—the system operator—therefore manages both.

[4] The steering committee had 14 members, representing the most important stakeholders.

However, the political process in California had foreclosed this possibility. The utilities had successfully lobbied the government to retain ownership of their transmission systems. The new system operator would therefore manage utilities' property rather than a public infrastructure. A coalition of industrial users and independent power producers was worried that this would compromise the system operator's neutrality. To ensure that the system operator would not rig the markets in the utilities' favor, they demanded that the financial markets be separated from grid management.[5]

From the perspective of market designers, however, such an organizational bifurcation had no discernible upside. It would simply make it more difficult to achieve coordination between the two sides. Prominent market designers contested the separation provision as soon as it entered the debate (Garber, Hogan, and Ruff 1994; Hogan 1994; Stoft 1996). As William Hogan at Harvard pointed out: "When we try to separate the auctions, we have a difficult and complex coordination problem. Nobody has been able to demonstrate a workable mechanism for coordinating such a complex process" (Hogan 1995, p. 30).[6] But politicians ignored these warnings because they could not otherwise gain the political support needed to move forward.

Designers did not view the separation provision as an insurmountable challenge. They planned to create an auction house for short-term financial contracts, the Power Exchange (PX), and an independent system operator. The steering committee declared it a priority to "provide a structure that will achieve effective coordination between the closely related functions of the independent system operator and the power exchange."[7]

However, the need for this bifurcation made system failures a real concern for the steering committee.[8] System failures occur when the interactions among system components become too complex for designers, lead to unintended interactions, and produce gaming incentives that derail the

[5] "California Energy Commission Comments on the Memorandum of Understanding," CPUC, October 2, 1995, box 19, folder 1, p. 5.

[6] In conjunction with San Diego Gas and Electric, Hogan even launched a last-ditch effort to convince FERC of an integrated design. San Diego Gas and Electric, July 3, 1996, "Notice of San Diego Gas & Electric of Withdrawal of Certain Pleadings," FERC ER96-1663.

[7] Pacific Gas and Electric Company, San Diego Gas and Electric Company, and Southern California Edison Company, August 1, 1996, "Official Transcript of Proceedings Vol 1," FERC ER96-1663, p. 105.

[8] Secondary literature has sometimes viewed WEPEX as doomed from the start (Walsh 2004; Isser 2015). This interpretation is consistent with sociological work where market design expertise appears as a rhetorical tool or epistemic framework for political negotiations (MacKenzie 2010; Breslau 2011; Reverdy and Breslau 2019). Yet, none of the design problems that led to illegal arbitrage games were predetermined by the political work. The games also did not always occur along the organizational separation between PX and ISO. Instead, they occurred between the ISO's imbalance market and the different financial markets within and beyond the ISO.

system. Because the designers now had to coordinate across organizational boundaries, such unintended interaction effects became a serious possibility.[9] As the literature would expect, modularization quickly became an attractive option to deal with this challenge. It allowed WEPEX to gradually disentangle which submarkets were related to which aspects of grid management and which parts of the system would be independent from each other. They could now develop integrated solutions for related modules even if these were ultimately administered by different organizations. Conversely, independent modules could be developed by teams that did not coordinate with each other. It was not the organizational separation between the system operator and the PX that dictated the modular structure of the system, but the functional relationships between the different aspects of markets and grid management. WEPEX could therefore build the systems for the system operator and the power exchange as an *integrated* system of loosely connected modules. "The [CA]ISO and PX will each consist of a number of subsystems," a trustee summarized the approach. Conceptually, "all of these systems were built so they could be procured separately or in an integrated package."[10]

Because the steering committee had now defined clear input/output relationships between related modules and demarcated independent modules, the system could be built by relatively small teams of experts. These experts could focus on the complexities they faced in their respective modules and ignore much of the rest of the system. They only had to cooperate with teams working on related modules. Toward the end of 1997 there were 77 separate teams working on different modules and subsystems that composed the larger architecture of CAISO alone.[11] Responsibilities were fluid and team composition could change quickly if the tasks demanded it. To coordinate with other teams who worked on related modules, each team had members responsible for "definition" or "implementation." Those working on definitions kept track of decisions and communicated them to related teams; the other members implemented the decisions.[12] This ensured that work on related tasks fitted together. Since many teams worked in parallel, large tables described deliverables and specified when what functionality had to be implemented. Project managers coordinated between teams and checked the

[9] Pacific Gas and Electric Company, San Diego Gas and Electric Company, and Southern California Edison Company, August 1, 1996, "Official Transcript of Proceedings Vol 3," FERC ER96-1663, p. 132.

[10] Direct Access Working Group, January 9, 1997, "Workshop on Communication and Data Systems" CPUC, box 30, folder 17.

[11] CAISO Management, 1997, "Phase II Work Item Status," CSA, ISO Board of Governors and Committee Meeting Files, box 6, folder 5 (hereafter, Phase II Status).

[12] S. David Freeman, February 25, 1997, "Testimony of S. David Freeman Before the California Senate Energy Committee," FERC ER96-1663, p. 3.

relationships between the different modules.[13] Beyond the project managers, there was little reporting to do. The hierarchy could be flat because the design rules of the system clearly specified who needed to coordinate with whom. This resonated with Silicon Valley's startup culture that early hires brought to the project. The steering committee hired technical experts who were willing to show initiative. "We specifically looked for people that were kind of driven crazy by bureaucracy," as the vice president of human resources put it (cited in O'Donnell 2003). Every few weeks, the steering committee met and reviewed the teams' work. In the second half of 1997, the executive level hired specialized vendors that collaborated with the teams to write the software and set up the hardware.[14]

In sum, the steering committee planned the auction markets as a system of loosely coupled modules that designers could build and manage in a corresponding division of labor. This not only helped to break down the coordination challenges WEPEX faced due to the separation provision. As in air traffic control systems (Vaughan 2021) and large infrastructure projects (Hughes 1998), the modular architecture allowed flexible teams to focus on smaller parts of the larger system but also ensured that teams carefully considered the impact of their decisions on related modules.[15] Given the vast complexity of their task and the intricate connections between grid management and markets, modularization seemed like an obvious organizational choice. And yet the system became vulnerable to persistent forms of gaming as soon as the markets opened. The next section reconstructs the nature of system failure to orient that analysis of why the modular design process went awry.

[13] For instance, the steering committee defined modules for "scheduling infrastructure," the "energy management system," and "scheduling applications." WEPEX Steering Committee, January 8, 1996–April 2, 1997. "Minutes." ArchiveX: https://web.archive.org/web/20000817212105/http://www.energyonline.com/wepex/, accessed March 10, 2021. The "scheduling infrastructure" processed results from markets inside and outside the system operator. This module thus had functional relationships with modules in the PX and the system operator. Teams coordinated across what would become the boundary between the two organizations. In contrast, the module "scheduling applications" only interacted with the system operators' internal scheduling infrastructure. These modules were developed by teams who needed no direct contact with those working on the financial markets. WEPEX, "Report on January 9th 1997 Direct Access Working Group Workshop," CPUC, box 30, folder 17, p. 17.

[14] ABB and subcontractor Ernest & Young built the software for ISO system; IM/HandEl of Norway for the PX. Perrot systems was responsible for integrating the different components in collaboration with the ISO staff. Phase II Status.

[15] In a progress report to the CPUC in 1997, the steering committee was optimistic that they had resolved the problems of market separation and would be able to bring the system online by the anticipated deadline. Phase II Status, p. 67.

Inconsistencies in the Platform Architecture Led to System Failures

As mandated by the separation provision, California's new system had two basic components: financial wholesale markets and a system operator. In the financial markets, buyers and sellers traded obligations for the delivery of energy at specific locations. So-called scheduling coordinators operated the different markets. The most important scheduling coordinator was the PX. It ran two main forward markets for electricity. The day-ahead market traded energy for delivery during the next day and the day-of market allowed traders to adjust their positions from the previous day. After the conclusion of trading, the auction software translated the results into schedules of anticipated generation and consumption and submitted them to CAISO.

CAISO was responsible for managing the actual flow of electrons on the grid. Its operators took the market results and tried to implement them as closely as possible while adjusting the schedules to the true operating conditions of the grid. To deal with deviations from the market schedules, the operators used several different markets of their own. They purchased energy to adjust for real-time fluctuations in a so-called imbalance or real-time market. In addition, they bought a variety of reliability services in "ancillary service markets." While operators could directly adjust generators' outputs through the imbalance markets, ancillary markets only sold backup capacity. For example, if a company sold "spinning reserves," they had to hold back some capacity that could come online within 10 minutes of being called. If called on, the electricity would then be sold separately. Figure 3 summarizes the structure of the system.

The financial markets were supposed to operate separately from the imbalance market. CAISO's operators used the imbalance markets to adjust

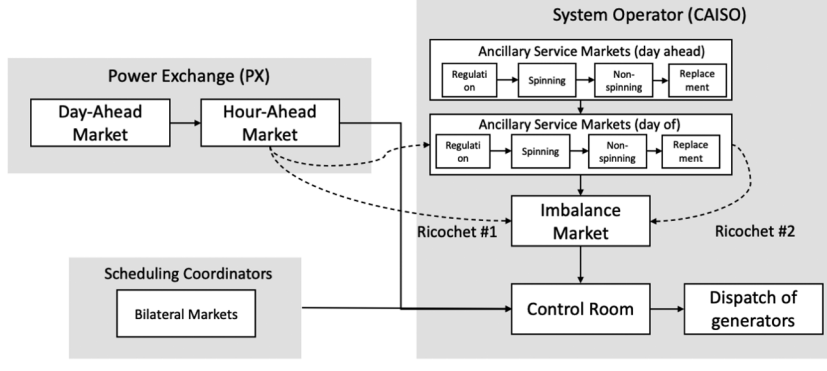


FIG. 3.—Structure of California's market system. The financial markets trade generic financial energy products; the imbalance markets trade physical resources under the control of the seller. Power marketers were prohibited from entering the imbalance markets. Pathways of illegal transactions are marked as dotted lines.

the physical system to real-time fluctuations. For this task, they needed to know the resources' technical characteristics in some detail. The sellers in these markets also needed to have direct control over the generators so that they could quickly follow instructions from the control room. In contrast, the financial markets traded generic obligations for future acts of production and consumption. The offers could be relatively unspecific and might summarize a portfolio of different resources into a single bid. The financial markets therefore allowed speculative moves that were unacceptable in the imbalance market. For example, traders could sell obligations to produce energy that were not backed by generation. In short, the financial markets were blunt planning instruments; the imbalance market was a tool to keep a vastly complex, technical system in balance.

Because the designers wanted the markets to sustain different activities, they imposed access restrictions on the imbalance market. Only companies who owned their generation assets, had direct control over these resources, and served real demand were allowed access. These restrictions generally excluded "power marketers," who made money via arbitrage trades, that is, by speculating on price differences between locations and time points. The market was supposed to operate as a separate module from the financial markets. Actions in one market were not supposed to affect actions in the other.

But throughout the existence of this system and two years before the crisis began, sellers of energy deployed various games to circumvent the access restrictions and thereby derail the market mechanism in both.[16] I will now consider these games from the perspective of the perpetrators to explain their rationale. I will then take the perspective of the designers and ask why these games were possible and profitable in the first place.

To give just one example, consider the so-called Ricochet strategy. Because most of California's neighboring states had not restructured their systems, CAISO assumed that energy imports were automatically under the control of the seller and did not impose restrictions. Companies like Enron, but also less likely suspects like the Los Angeles Department of Water Resources, colluded with each other to take advantage of this loophole (McNamara 2002; Navarro and Shames 2003; Kleit 2018; Nix, Decker, and Wolf 2021). They would buy energy in California and then schedule this energy

[16] Although Enron initially developed these games, practically *all* sellers of energy played them throughout the existence of these markets, many starting a full two years before the energy crisis. The FERC initially defined manipulation on the basis of the "Enron memos" in the aftermath of the crisis. It therefore did not become apparent how pervasive and general these strategies were in the market. This changed only after a remand from the Ninth Circuit Court of Appeals, which allowed buyers of energy to file thousands of exhibits that demonstrated a variety of additional violations to California's tariffs between 2012 and 2013 (Taylor 2015).

for export to another party outside California. This party sold the power back to the power marketer for a small "parking fee." Since it now seemed like the energy was coming from "outside" California, the power marketer could sell the energy as an import to CAISO and circumvent the control requirements of the tariff. The imports looked like they were coming from other utilities, so it seemed like they came from generators under the control of the seller. This way, the power marketers could access the market and realize profits from price differences. Enron's traders called this strategy "Ricochet" because the energy would bump around the (fictional) network model before "returning" to California in a different guise.

Far from simply arbitraging price differences, the strategy derailed the market design and impacted the reliability of the grid. First, the transactions in the financial markets no longer reflected attempts to trade obligations for the future delivery of energy. Instead, they were designed to obfuscate the true origin of electricity. This affected incentives for other actors and derailed the market mechanism. Second, the contracts that ended up with the system operator might not meet the technical requirements. For example, if a trade represented a short sale, that is, a promise to buy the corresponding energy at the moment of delivery, the power marketer might not be able to realize this promise. The bid would simply disappear when the operators tried to draw on it. If the other resources did not suffice to clear the market, the remaining sellers gained market power. In short, Ricochet undermined the market design by linking two parts of the system that were meant to be separate. Related strategies achieved the same result via different pathways.[17]

So far, I have discussed illegal arbitrage from the perspective of the perpetrators. But to understand why these games represent system failures, we need to consider them from the perspective of the designers. What elements of the market design first created the incentives and opportunities for this game? Ricochet was possible because of the flawed import/export rules. But it was profitable because market actors perceived systematic price differences between CAISO's imbalance market and the financial markets.

In principle, there should not have been such differences. The imbalance markets relied on information that only the grid operators had in the minutes before dispatch. Any price differences should therefore have been temporary and random. But this was not the case. Instead, *administrative* differences between the market interfaces translated into systematic price differences.

[17] FERC initially misinterpreted Ricochet as a limited strategy to sell energy "out-of-market" to CAISO. Such sales were not subject to price caps. However, subsequent litigation established that Ricochet was part of a suite of strategies that tried to arbitrage price differences between the financial and the imbalance markets (Taylor et al. 2015). CAISO Department of Market Analysis, October 4, 2002, Analysis of Trading and Scheduling Strategies Described in "Enron Memos," FERC PA02-02, pp. 28–30.

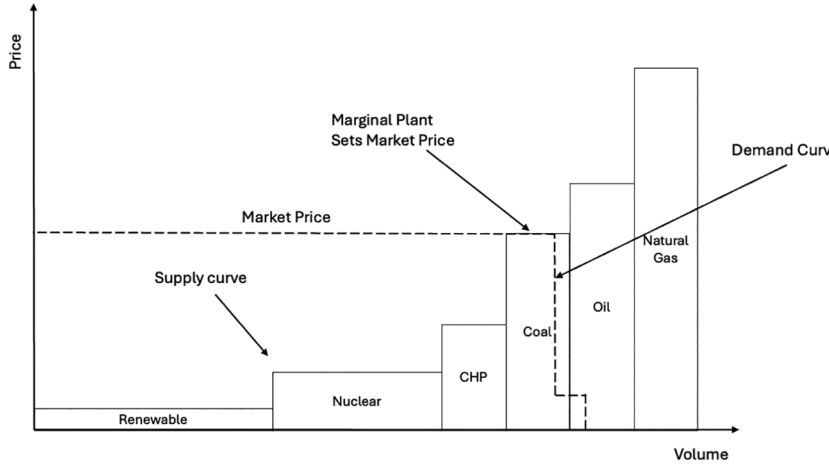


FIG. 4.—Illustration of merit order dispatch (as used in PX day-ahead auction). The calculation of both curves can vary depending on the information about the physical system that is included in the curves' construction.

In each case, the designers adopted slightly different rules, protocols, and interfaces for the imbalance market than for the different financial markets.

For instance, the imbalance and the financial markets used different protocols to determine winning bids.[18] To enter the financial markets, buyers and sellers submitted bids to sell or buy energy.[19] The auction software aggregated all bids into global supply and demand curves. Then it set a uniform market clearing price at the price of the last megawatt hour necessary to meet demand. By ordering bids and intersecting the two curves, the software found the so-called merit order dispatch—the cheapest combination of generators to serve demand in a given hour, relative to various reliability constraints. Figure 4 illustrates the idea. The global supply curve consists of generators whose marginal costs increase with output, forming an ascending step function. The inelastic demand curve on the right intersects the supply curve at the last generator necessary to meet demand. This marginal generator sets the market price. Generators below the dotted line sell their

[18] Throughout the energy crisis of 2000–2001, the financial markets used different price caps than the imbalance markets. That is, the financial markets set different limits on the highest possible price in the market. This impacted bidding behavior under conditions of limited supply and led to divergent prices.

[19] If constraints on California's main transmission lines would block the implementation of this ideal dispatch order, an implicit market for transmission rights adjusted the schedules to incorporate these constraints. James G. Kritikson, February 2000, "California Electricity Market Primer, prepared for the CalPX Board of Governors," CSA, Electricity Oversight Board Subject Files, box 18, folder 12.

energy and profit to the extent that their costs are lower than the market price.

The imbalance markets at CAISO operated differently. Whenever a forward market concluded, the scheduling coordinators gave the winners the opportunity to submit adjustment bids. Such bids were offers to increment ("inc") or decrement ("dec") the output of generators that were scheduled to deliver energy. The bids were also submitted as supply curves, indicating readiness to provide or reduce different quantities of electricity at different prices. Just like the financial markets, CAISO's software constructed a supply curve from these bids (as in fig. 4). But there was no corresponding demand curve. Instead of intersecting two curves automatically, CAISO's operators received the supply curve as a stack of resources. They then drew on the resources from this stack as needed. This had two consequences.

First, operators could skip resources in the bid stack and sometimes did skip cheaper for more expensive resources. Second, the software separated inc and dec bids into two stacks and thus nudged the operators to dispatch them separately—rather than simply identifying the optimal combination of generator outputs.[20] The market price would then be determined by the cost of the last resource used to adjust the system in a 10-minute interval. This generated different supply and demand curves than the financial markets, and at the intersection of supply and demand, the imbalance market would now produce a different price.

There were multiple other differences along similar lines, effectively changing the way the supply curve was calculated in the different markets: operational rules about how to deal with exceptions, price caps, priority rules for different resources, differences in the way the complexity of the physical electricity was represented in the market, and so on.[21] Such differences did not just exist between the imbalance and the financial markets but also between the ancillary service markets and the imbalance market, as well as two closely related markets for congestion management. As soon as market actors became aware of these formal differences and found ways to circumvent any restrictions on the corresponding trades, they could make substantial profits with riskless arbitrage transactions. As one of Enron's coconspirators put it in a letter: "An overabundance of strategy categories

[20] CAISO Market Surveillance Unit, June 1999, "Annual Report on Market Issues and Performance," http://www.caiso.com/market/Pages/MarketMonitoring/MarketMonitoringArchive/Default.aspx, accessed March 10, 2021 (hereafter, CAISO Annual Report).

[21] CAISO's Market Monitoring Committee investigated how different dispatch practices could produce price differentials that would lead to deviant behavior by market participants. A good summary of the findings from 1998 and 1999 can be found in Frank A. Wolak, October 18, 1999, "MSC Report on Redesign of California Real-Time Energy and Ancillary Services Markets," FERC ER98-2834, pp. 97–100.

exist; ranging from just playing 'gaps' in the protocols, to taking advantage of self-created congestion."[22]

Manipulative games like Ricochet responded to inconsistencies in the rules, procedures, and interfaces between different submarkets. These inconsistencies generated incentives and opportunities for behavior that violated the market design. They were system failures because they originated in inconsistencies between rules, procedures, and interfaces that WEPEX failed to control. This, then, leads to the questions for the remainder of the article: Why did market designers adopt these inconsistent rules, why did intermediaries fail to catch the problem, and why did they not manage to resolve the problem after the markets opened in 1998?

MARKET DESIGN AND THE PERILS OF MODULARIZATION

The Problem of Incomplete Coordination

By zooming in on the work of teams that built the software for the imbalance and the financial markets, we can see that inconsistencies emerged when the teams translated abstract market design blueprints into a concrete technology. While this is a fairly universal problem of design work (Stinchcombe 2001), the case shows that it assumes a much greater significance in market design than in safety-critical organizations. Modularization requires that *employees* collaborate to maintain the relationship between modules. But the relationship between platform markets depends partly on the activities of *market actors*. They do not respect the intended relationship between modules if it is not in their interest and search for inconsistencies anywhere in the system. This means that problematic inconsistencies can appear anywhere, undermining the guiding principle behind modularization.

A single team of engineers, economists, and external programmers working for the software vendor ABB implemented the software for the imbalance market. Operations researchers Vikram Budhraja and Jack Allen, together with engineer Ziad Alaywan, led this work.[23] They implemented the imbalance markets in a program called BEEP, Balancing and Ex Post Energy Pricing.[24] When the designers developed BEEP, they followed the technical prescriptions of a "tariff"—a document that provided general directives for the work of the teams and spelled out how the different parts of the system were related to each other. The tariff stated that "the sources of Imbalance Energy . . . will be arranged in merit order of Energy bid prices,

[22] E. Smith, "Re: Real Time Market Monitoring and Optimization," Letter to Rich Davis Head of Enron Capital and Trade Resources Corp. SNO-86, FERC EL03-180, p. 1.

[23] Work schedules for the internal development of the market software report team leaders; cf. Phase II Status.

[24] CAISO, Annual Report, sec. 2.7.

with respect to both incremental and decremental Energy bids."[25] While merit order dispatch had both a formal engineering and an economic definition, the tariff did not specify exactly *how* incremental and decremental bids were to be displayed in the final merit order. The basic idea behind merit order dispatch is captured in figure 4—a stack of resources arranged in ascending order by marginal cost of production. But the concrete calculation of the stack (how to calculate the *cost* of generators relative to each other) and its display in the BEEP interface (how to represent these relations visually) were ambiguous.

This is not atypical. All formal blueprints are practically indeterminate at some level. The sociology of knowledge has long shown that contexts of work first make the requirements for implementation concrete (Stinchcombe 2001; Bailey and Leonardi 2015).[26] As the theory suggests, the design team interpreted their task in terms of the overarching purpose of their module. As a report stated, the "applications were designed primarily to serve as tools to assist in the decision-making relating to the CAISO control area."[27]

CAISO's control room was the beating heart of the organization. Its layout was optimized to aid real-time system management. It featured a variety of semicircular control stations that were oriented to a large map board. The map provided a schematic overview of California's high-voltage electricity grid. Hundreds of small red and green lights represented every major generation facility and electric switchyard in California. LED digits indicated the amount of power flowing on each link. The workstations were dedicated to specialized tasks. Glancing up from their individual screens, operators could check the real-time status of the grid on the board. This way, the map focused the dispersed activities in the large room and provided a common reference point. The BEEP software ran on a dedicated workstation where operators constantly adjusted generators' outputs to stabilize the electricity system.

As in financial trading operations (Beunza and Stark 2012), the team tried to adjust the interface of the BEEP software for the specific purpose in the control room. To use the BEEP stack quickly and intuitively, the operators needed to be able to draw on the resources with some discretion. As auditors of the design process summarized: "The BEEP system is also automated; however, its execution involves manual procedures and human judgment,

[25] CAISO Management, April 1998, "ISO Scheduling Protocol, CAISO Tariff," FERC EC96-19-017, sec. 11.2.

[26] This goes back to an old insight about the pragmatic nature of rule following (Wittgenstein 2010).

[27] CAISO Management, 1998, "CAISO Report on Management Assertion, Relating to Ancillary Services, Management, Balancing Energy, and Ex-Post Pricing, and Congestion Management Systems." CSA, ISO Board of Governors and Committee Meeting Files, box 6, folder 4, p. 12 (hereafter, Management Assertion).

applied primarily by control room dispatch operators, who determine which generation units are ultimately utilized to satisfy the CAISO real time energy needs."[28] The same thinking applied to the layout of the software. After trying several different configurations, the designers settled on an interface that separated bids into two stacks that operators could draw from.[29] Rather than representing a single list of generators with offers to increment or decrement, they presented one for offers to decrement and one for offers to increment. This made it easy for operators to see what inc and dec resources were available for the different tasks.[30]

This setup was not an invitation to violate merit order dispatch. Quite the contrary. It was organized around a different, local definition of this concept. Due to the information available in the control room, operators had a much more comprehensive view of the system than the actors in the financial markets. The financial markets dealt with generic contracts for the delivery of energy. The control room dealt with specific generators whose outputs into the vastly complex grid needed to be brought into alignment with each other. From operators' more encompassing perspective, it could be *reasonable* to skip resources or change the order in which they were dispatched—to preserve "true" merit order dispatch.[31] Unfortunately, these operational characteristics introduced the very inconsistencies that made games like Ricochet profitable.

In sum, then, the market designers implemented a local definition of how merit order dispatch should be calculated and displayed when they developed the BEEP system. This happened because the tariff left indeterminate what precisely it meant to implement this concept in the complex sociotechnical system of California's electricity system. Their implementation was a prudent response (it helped the work of the operator) to a local problem (how to organize the interfaces of the control room). Yet, their solution

[28] Price Waterhouse, May 27, 1998, "Report of Independent Accountants to Board of Governors," CSA, ISO Board of Governors and Committee Meeting Files, box 6, folder 4.

[29] Cf. Paul Gribik, July 22, 2002, "Consequences of the Imbalance Energy Market Protocols," submitted in "California's Electricity Market: The Case of Perot Systems," Subcommittee on Energy Policy, Natural Resources and Regulatory Affairs of the Committee on Government reform, House of Representatives, 107th Congress, 2nd Session, July 22, 2002, p. 207.

[30] CAISO Annual Report, sec. 4.1/2.

[31] E.g., operators might have to skip resources when they were scattered too far across the map, when they could only provide small amounts of energy, or when they should better provide valuable backup services. It could also make sense to skip cheaper for more expensive resources if the more expensive generators had "minimum run times," that is, had to run for several hours to be economical. These generators had very high startup costs but would become cheaper over time. PX Compliance Unit, November 7, 2000, "Price Movement in California's PX markets, May-September 2000," FERC ER96-1663, p. 82.

was inconsistent with the way the other design teams had implemented the same concept in the financial markets.

Particularly when designing new systems, such inconsistencies are a common "liability of technological innovation" (Stinchcombe 1965). Modular divisions of labor are therefore not normally vulnerable to this problem. Organizations that manage safety-critical technologies can usually discover such problems because *employees* collaborate to maintain the relationships between the modules. For instance, air traffic controllers communicate with other air traffic controllers about the definition of sectors in the sky. Airplanes cooperate with their instructions. If decisions of one team have adverse consequences for decisions of another team, they will alert each other and negotiate how to deal with the situation. Moreover, airplanes will make an active effort to communicate clearly how they are relating to the system of sectors. The cooperative "boundary struggles" are crucial to determine what aspects of a module are relevant to the workstation and which concern others as well (Vaughan 2021). Similarly, crews on flight desks are locked into silent negotiations about who can do what tasks most efficiently (Roberts 1990). These dynamics first stabilize the system's "design rules"—the input/output relationships between different parts of the system.

In market design, problematic inconsistencies between rules, procedures, and interfaces first turn into problems when *market actors* identify and act on them. It thus depends on the perspective of the market actors how exactly the markets are related to each other. But market actors look for profitable trades *anywhere* in the system. This means that California's designers had no good indication of where to look for problematic inconsistencies.

Indeed, they had reasons *not* to look at the relationship between the imbalance market and the different financial markets. The platform architecture stipulated that the BEEP module would be independent from scheduling coordinators. The various financial markets submitted their results to CAISO's scheduling infrastructure, which passed the information to the module "scheduling applications" from where BEEP retrieved it.[32] The BEEP module thus had no direct relationship to the financial markets. The modular division of labor suggested that they could ignore the operation of these markets for the purpose of their module.[33]

[32] Management Assertion.

[33] Reports about system integration suggest that they did indeed treat other modules as black boxes if the design rules specified them as such. For example, the ISO ran 50 extended system tests to see whether all parts of the system interacted correctly. They invited market participants to test the trading system and taught them how to use the software. Throughout these tests, the BEEP team only examined whether the data arrived in the correct format from the scheduling infrastructure. Working with random data and users who followed the rules, the team's main concern was to get the input/output relations with the scheduling infrastructure right. While the tests led to amendments to the

Even if market designers knew that gaming might create links between otherwise unrelated parts of the system architecture, they did not know where exactly to look. In 1996, the team working on bidding rules for the new auctions in the PX hired the market designer Robert Wilson from Stanford and the experimentalist Charles Plott at CalTech. As early as 1997, Wilson filed a brief with the WEPEX steering committee, noting that they needed to control arbitrage between the different components of the system: "A central requirement in the overall design of the energy markets is enabling sufficient arbitrage."[34] Wilson was aware that lack of coordination between the auctions might produce problems like Ricochet.

But while he saw the problem, he could not consider all possibilities. His main concern was the relationship between scheduling coordinators rather than the relationship between them and the imbalance markets. He suggested that one might solve the problem by "enabling a market for trading Inc/Dec options among the various scheduling coordinators, including the PX."[35] Because this was not his job, Wilson merely noted the problem as an aside and moved on.[36] Not only did the problem fall between the cracks of teams' responsibilities. Even if it had been resolved, it would not have affected the relationship between the scheduling coordinators and the imbalance market. There were simply too many ways in which the actions of market actors could render the preordained interface relations between modules problematic. Presumably for the same reason, legal counsel did not consider the impact of import/export rules for the dynamics in California's internal markets, creating the opportunities for Ricochet.

In sum, designers followed local considerations to concretize abstract concepts from the tariff. Decisions were typically prudent responses (helping operators) to local problems (how to organize the control room). Yet, due to the "liability of technological innovation," such local considerations produced subtle inconsistencies in the operation of the different auctions. Because the modular division of labor defined the corresponding decisions as purely local, it was not set up to allow designers to trace the consequences

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CAISO tariff, it did not evaluate the incentives in the interplay between imbalance and financial markets. Cf. CAISO Management, March 3, 1998, "Amendment No. 4 to the ISO Operating Agreement and Tariff, Including the ISO Protocols," FERC EC96-19017, pp. 6–8.

[34] Robert Wilson, March–May 1997, "Activity Rules for the Power Exchange and Priority Pricing of Ancillary Services, Reports to the California PX and ISO Trusts for Power Industry Restructuring," ArchiveX, https://web.archive.org/web/20000817212118/http://www.energyonline.com/wepex/reports/reports2.html, capture from February 4, 1997, last accessed October 28, 2024 (hereafter Activity Rules for Power Exchange).

[35] Activity Rules for the Power Exchange.

[36] Interestingly, he complained early and publicly about the distributed division of labor (Stoft 1996).

of these decisions elsewhere. Indeed, if market actors had simply followed the rules of the markets, there would not have been a problem. But market actors did not respect the relationships between the modules and drew on information anywhere in the system. This made it very difficult for designers to identify with which teams they needed to coordinate to avoid problematic inconsistencies. But why did the intermediaries who managed the relationship between teams not identify these problems?

Structural Secrecy in Market Design

Since no modularization is perfect, organizations do not completely seal off their teams from each other. Most design projects have intermediary and executive levels, on which managers examine the interface relations between subsystems and coordinate between teams (Hughes 1998). They analyze the relations between teams and update the design rules in response to new interdependencies. Aware of problems that might emerge between the two sides of the system, the WEPEX steering committee constantly looked for problems like Ricochet and hired economists and engineers who analyzed the protocols for the new markets.

However, such intermediaries always consider problems at a certain level of *abstraction*. They focus on decisions that are related to problems in the interplay between modules in the larger system. This leaves a variety of supposedly local decisions that tend to produce problematic incentives in relation to other modules. To discover these problems, intermediaries would need to understand the local context fully and then evaluate its relation to the larger system. This is not their job. To the extent that the modular division of labor guides intermediaries' attention, it structurally obscures problematic inconsistencies elsewhere in the system.

The decisions about the layout of the BEEP stack are a good example, because the relationship between the PX and the ISO was a topic of explicit concern at the steering committee. As outlined before, the decision to split California's system into two separate organizations had been a political compromise. While designers were confident that they could accomplish the separation, implementing it presented a variety of technical and conceptual challenges. In a comment, a prominent market designer wrote, "The difficulty is not with power exchanges separate from the ISO per se, the difficulty is in the definition of the role of the ISO and the connection with the power exchanges."[37] It was a high-level, conceptual problem, and the members of the steering committee spent much time pondering it.

[37] San Diego Gas and Electric Company, 1995, "Comments of San Diego Gas & Electric Company to Coordinating Commissioner Ruling," CPUC, box 19, folder 49a.

In the initial application for authority to establish the new system, the steering committee already pinpointed the basic problem at the heart of Ricochet. They noted that "the PX cannot function effectively without the ISO, since the PX bidders depend on the ISO for real-time balancing of load and generation resources." Correctly, they inferred that access restrictions on the imbalance markets could "only work if there are no substantial price differences" between the markets.[38] And they quickly settled on a solution that seemed to solve the problem. In 1996, they determined that the financial markets "will establish an initial merit order dispatch for each hour in the next day," and in the next step, "the ISO redispatches PX loads and resources in merit order . . . based on incremental/decremental prices."[39] Both the PX and the ISO were to clear the market on the basis of merit order dispatch, that is, the cheapest combination of generator outputs to meet demand in a given hour, relative to system constraints. With consistent market clearing protocols, the only differences between the two markets would be random fluctuations in price on exceedingly small quantities of "physical capacity." This decision shaped the directive of the tariff, which was then handed to the dispatch integration team in the ISO. From the global perspective of the Steering Committee, these two parts of the system were now truly separate and could be developed independently from each other. Accordingly, intermediaries mainly examined whether the design teams had put rules into place that were consistent with merit order dispatch. As we saw, CAISO's implementation did not violate the directive. But due to the practical indeterminacy of the concept, the teams developed divergent interpretations of merit order dispatch. Because these interpretations developed in response to highly local issues and because the modules were presumed to be independent by virtue of the more general definition of merit order dispatch, the intermediaries ignored them. They therefore missed the inconsistencies that had already created the incentives for market actors to link these modules and thereby violate the design rules of the system.

Diane Vaughan has coined the term "structural secrecy" to describe situations where the reporting architectures of organizations block signals of problems from arriving at relevant locations (Vaughan 2008). This describes the problem well, although market design added a twist. In organizations that manage risky technologies, structural secrecy is a by-product of problematic work practices inside the organization. For example, the gradual normalization of deviant practices means that important signals of danger are

[38] Pacific Gas and Electric, San Diego Gas and Electric, and Southern California Edison, April 29, 1996, "Joint Application of PG&E, SDG&E, and SCE for Authority to Sell Electric Energy at Market-Based Rates Using a Power Exchange," FERC ER96-1663, p. 5.

[39] WEPEX Steering Committee. February 20–21, 1996. "Minutes," ArchiveX https://web.archive.org/web/20000817212105/http://www.energyonline.com/wepex/, last accessed October 28, 2024.

blunted before they can reach the relevant decision-maker (Starbuck and Milliken 1988; Vaughan [1996] 2016). In market design, structural secrecy in the organization managing the market is the result of adaptive behavior by market actors. Market actors are actively looking for loopholes and inconsistencies that emerge in the coordination gaps between design teams. Once they exploit these gaps, the de facto operation of the system changes. Parts of the system become interrelated that are supposed to be separate. The existing division of labor among the designers then no longer mirrors the system architecture, and relevant information does not reach those who would need it. Structural secrecy thus results from the way the adaptive behavior of antagonistic market actors destabilizes the presumed relations between market modules.[40]

Temporal Instability

A third problem explains why it was difficult to fix the system failure ex post—that is, once the system became operational and designers began to discover games like Ricochet. Academic market designers frequently assume that designers can iteratively discover and address global misalignments between rules and thus resolve market design failures over time (Wilson 1998; Börgers 2015). However, the logic of modularization makes such a process of iterative perfection unlikely. Local processes of creative problem solving occur concomitantly. Designers in different teams will continuously and experimentally tweak rules in response to local problems. Each time they change a rule, the global baseline of incentives for market participants' behavior changes. To the extent that this leads to new behavior, old fixes can interact with new ones to produce incentives for gaming. This means that iterative problem solving in different modules does not converge on a globally consistent set of rules.

Precisely this problem blocked California's designers from eradicating the incentives behind Ricochet. The new system became operational in 1998. At that point, CAISO and PX had become independent organizational entities with more traditional, bureaucratic structures (Roe and Schulman 2008). However, to diagnose problems and implement design changes, they still relied on flat hierarchies and flexible teams. When problems emerged, the management assigned specialized teams to develop solutions.[41] The PX

[40] From the perspective of market actors, the games represent complex secrets. The activities are designed to fall in the gaps of regulatory attention, effectively allowing these actors to hide in plain sight (Rilinger 2019).

[41] Technical Advisory Committee Meeting, November 11, 1998, "Memo: Negative Supplemental Energy Pricing," CSA, ISO Board of Governors and Committee Meeting Files, box 6, folder 6.

proceeded similarly.[42] Throughout the design and redesign process, the steering committee hired market design experts who worked for Perot Systems, London Analytics, and Price Waterhouse. They actively searched for ways in which protocols might lead to gaming activities. After the markets started, market monitoring units continued this work. But these efforts to identify and resolve inconsistencies between rules in different parts of the system did not push the system toward perfection. Instead, new games and new versions of old games kept emerging. In the first month of market operation, the ISO filed seven amendments to its tariff. By the end of 1999, they had filed 23 major tariff changes. Many amendments contained references to games that had to be eradicated via rule changes.

The basic problem was this: Even if designers detected a problem in the interplay between different rules, adjustments necessarily occurred on the local level. Since multiple reactions occurred concomitantly, it was never possible to account for all interactions between components on the global level. A market designer described the work process in an interview from 1999: "There was a lot of work, a lot of fast changes, a lot of reactions—as soon as [one issue] was done, we started the next one" (O'Donnell 2003, p. 115). The fast reactions required that designers relied on experimentalism. As one engineer remembers: "We tried in the design to deal with mismatches and holes the best we could, but the thought was that we were going to do the best we could. And then if something came up, we'd change it."[43] Designers would identify problems, develop a quick local fix, and then move on to the next issue. But because changes happened everywhere, the global baseline of incentives kept changing and thus the nature of the problems the rule changes were meant to address. Worse, market actors intentionally looked for combinations of rules that intermediaries had not checked yet. Since the behavioral foundation of the system was not stable, iterative attempts to identify problems and solve them via rule changes were unending. Solutions to old problems could suddenly produce new problems as fixes in other parts of the system affected the global balance of incentives.

Again, Ricochet provides a useful illustration because attempts to fix this game interacted with an older fix and merely shifted the problem to a different venue (Ricochet 2, in fig. 3). One of the most persistent problems during the first year of operation occurred in the ancillary service markets—these markets traded four different kinds of backup service that could be traded in day-ahead and real-time markets. Services that come online faster are generally considered higher quality than those that are slower because they can be substituted for slower resources. CAISO had inflexible needs for each

[42] E.g., PX Technical Committee, November 11, 1998, "Minutes" CSA, Power Exchange Corporate Meeting Files 1998–2006, box 13, folder 13.

[43] Phone interview with CAISO engineer, November 22, 2017.

of these services and had separate auction markets for them. When there was not enough supply in one of the markets, sellers could drive up prices without fear of losing business. This created the bizarre situation where lower-quality resources could command much higher prices than higher-quality resources. On July 9, 1998, the market for the inferior "replacement reserves" spiked to $9,999—the maximal input value on the software interface—even though there were ample supplies of higher-quality services.

To solve this problem, a special task force under the leadership of Stanford economists restructured the entire ancillary service market system. Forced by the complexity of the system to take a narrow perspective, they (initially) presumed that the rest of the system remained fixed and focused on the relationship between the ancillary markets. Apart from new market mechanisms that enabled the ISO to substitute higher-quality for lower-quality services, the ISO moved 7% of its demand from the day-ahead market to the day-of market. This was supposed to depress the price in the day-ahead markets. The price for the whole market was set at the price of the last generator necessary to meet demand. The less demand there was in the market, the lower the price would therefore be. Indeed, the cost of ancillary services declined by roughly 50% from April 1998 to March 1999.[44]

However, operators now bought substantial amounts of ancillary services so close to dispatch that there was limited time to verify the resources or substitute them for alternatives (day-of ancillary service markets in fig. 3). This created a latent problem. Just like the imbalance markets, the ancillary service markets were supposed to be independent from the financial markets at the PX. But there was a link between the ancillary services markets and the imbalance markets. If operators did not need the ancillary services, they could select incs or decs from generators that had previously sold their capacity as ancillary service. In that case, the generator got paid twice: for providing the backup capacity in the ancillary markets and for providing the energy in the imbalance market.

The potential to get paid twice could amplify the price differences between the PX markets and the CAISO markets substantially. Such mechanical price differences might, once again, trigger a game like Ricochet. Benefiting from the lax controls in the real-time ancillary markets, the market participants might illegally move imports into the real-time ancillary markets and then shift to the imbalance markets (dashed arrows for Ricochet 2 in fig. 3).

However, at first this did not happen because capacity in the ancillary service markets was cheap, and operators usually skipped these resources

[44] CAISO Market Surveillance Committee, October 18, 1999, "Report on Redesign of California Real-Time Energy and Ancillary Services Markets," CSA, ISO Market Surveillance Committee Meeting Files, box 13, folder 4.

to keep them available for reliability purposes. Thus, there was no good way to move high-priced bids from the ancillary markets to the imbalance markets. Because the path was blocked, power marketers played the standard version of Ricochet and left these markets alone. However, the 7% rule eventually realized the inherent risk of this setup.

Ironically, this happened when the designers addressed the original Ricochet strategy—without thinking about the relationship between the rules they changed and those that structured the relationship between the ancillary service markets. During the energy crisis, CAISO discovered the problem. Because Ricochet relied on false imports, a local team from the market monitoring unit implemented a new rule to limit out-of-market purchases for imbalance. Again, they considered the problem in isolation—they saw that imports created problems and increased the barriers. But the change reverberated throughout the system as a whole. To compensate for the loss of imports, the ISO needed to increase its purchase of incs and decs from ancillary services. Thus, it became more likely that energy bid into the real-time ancillary service market would be sold in the imbalance markets (solid arrow, fig. 3).

Because the ISO had moved a large amount of its demand for ancillary services into the real-time market (the 7% rule) and would draw on them to provide imbalance energy, power marketers now had an incentive to circumvent the access restrictions and enter imbalance markets via the real-time ancillary service markets. They would use the same old import-export scheme to enter the ancillary markets—imports were still acceptable here—and enjoy getting paid twice when the operators had to use these resources. Shortly after the rule change took effect, a report diagnosed that "it appears the purchase of significant amounts of Replacement Reserve . . . may have created an additional incentive for suppliers . . . to shift additional capacity into the Ancillary Services . . . and real time markets."[45]

What used to be a solution to a behavioral problem (shifting demand into real-time ancillary markets) created a new problem when a rule in another part of the system shifted (restrictions of imports for imbalance energy). The new rule changed the configuration of incentives that actors faced in the old market (large potential arbitrage profits between financial and ancillary service plus imbalance market), leading to different behavior (gaming the imbalance market via the ancillary market) that turned the old solution into a source of new problems.

At the heart of the matter was a basic problem: The management of electricity systems was technically demanding and complex. Teams working on individual parts constantly confronted new, small problems. This required

[45] CAISO Department of Market Analysis, August 10, 2000, "Report on California Energy Market Issues and Performance: May-June, 2000," FERC PA02-2-000.

local adaptations whose consequences should have been limited to the individual modules—under the assumption that the interface relations between the modules were fixed. That is, adaptations were based on a targeted search for solutions given the assumption that the rest of the system could be treated as constant. But since market actors could relate each part of the system to any other in their search for arbitrage opportunities, every adjustment had the potential to shift the balance of incentives and lead to new types of behavior that derail the design. This changed the baseline against which the designers needed to define solutions and rendered old solutions potentially ineffective. This made market design extremely challenging. The very technique that designers used to make the complexity of the system tractable produced the system failures they were trying to avoid.[46]

SCOPE CONDITIONS

A frequent refrain in the literature on safety in organizations is that experts at the "coalface" of the organization are best able to manage complex technologies. Here it is easiest to react quickly and appropriately to surprising interactions among system elements (Highsmith 2009; Huising and Silbey 2018; Vaughan 2021, 2023). A key to empowering frontline experts is to decompose complex technical systems into loosely connected modules. The experts can now manage the different components of the system in a flat hierarchy.

The empirical analysis of the California case suggests a general reason why this strategy may be tempting but not entirely appropriate for the design of digital platform markets. Modularization requires that organizational actors respect and actively maintain the system's design rules—the input-output relationships between the different modules. Organizations that manage safety-critical technologies deploy a variety of managerial tools to foster and develop this cooperative orientation—a unique "ethnocognition" (Vaughan 2021) that allows reliable operation in the face of system effects that threaten the interface relations between modules. They extensively train employees, foster a culture of "obsession with failures," and put into place organizational routines for reflexive learning (Bourrier 2011; Cantu et al. 2020).

In contrast, in market design, the relationship between the different markets depends partly on the activities of market actors. These actors are not members of the organization that operates the market and do not have any special allegiances to designers' goals. Designers coordinate these actors by structuring the rules, procedures, and interfaces that define their choice

[46] Indeed, this problem is closely analogous to Perrow's observation that the introduction of redundancies can make a system more rather than less prone to accidents because it increases the system's dynamic complexity (Perrow 1999).

environment. Not only can minor inconsistencies in the market rules, procedures, and interfaces lead actors to violate the relationships between the modules and deviate from the intended logic of action, but profit-oriented market actors can search for such inconsistencies anywhere. This undermines the logic of modularization because it becomes difficult for designers to know where problems might emerge. Prudent design decisions in one part of the system can produce gaming incentives in nominally unrelated parts of the system. Moreover, it is very difficult for designers to fix this problem iteratively because each decision can affect the global baseline of incentives in the system and produce problems elsewhere. In short, the California case suggests that modularization, although common and a priori reasonable, does not work as advertised for platform markets. Due to the structural antagonism of market actors, it is more likely to cause rather than to prevent system failures.

However, we should not be too quick to draw generalizations. The California case is unique and extreme in important respects. We must therefore consider the nature of and scope conditions for the theoretical argument of this article. First, California's auction markets represent a particularly challenging type of market design. These markets were meant to directly inform the grid management by operators in CAISO's control room. Very small deviations from the intended market logic could therefore have serious consequences. Not all market mechanisms have such tight margins of error. When there are many possible solutions to the underlying allocation problem, mechanisms tend to be robust to some degree of gaming (Carroll 2019). This is true for rideshare apps or auctions for digital advertisements. In these cases, designers may even choose to make trade-offs between robust incentives and goals like fairness or revenue maximization (Kominers et al. 2017).

Nonetheless, while such gaming does not necessarily lead to system failures, it is generally undesirable, and modularization is more likely to produce rather than prevent it. Given that it is a prominent technique of software design and modular divisions of labor are increasingly common in "postbureaucratic organizations" (Davis 2015), the argument in this article may explain why illegal arbitrage is so difficult to avoid on many financial platforms and power markets (Hao et al. 2023)—most recently in Texas's ERCOT markets, where bitcoin miners have developed a new business model around such arbitrage trades (Khalili 2024). It may also explain why so many gig platforms have difficulties in rooting out gaming despite their unprecedented control over, and insight into, the market process (Rahman 2021; Cameron and Rahman 2022). Conversely, it offers a reason why successful platform markets are often built and overseen by *centralized* organizations with long divisions of labor (Vergne 2020; Baldwin 2023). For instance, the PJM Interconnect operates electricity markets in the eastern

parts of the United States. For the first decade of the 21st century, PJM was the poster child for successful market design. A management committee at the top of the organization made most technical design decisions about the initial design in consultation with three other committees (Lambert 2001, pp. 55, 91, 99). However, such centralized organizational structures are not a panacea. They are less suitable for quick adaptation to volatile and uncertain environments. Centralized design may therefore have to be abandoned as the system becomes more complex. Incidentally, PJM has gradually shifted to a more distributed design approach that has recently led to serious failures (Hogan 2021; Simeone 2021). Yet, while markets are usually celebrated as the solution to centralized planning (Hayek 1948), planned markets may, somewhat ironically, require a centralized planner.

Second, California's markets had weak, fragmented, and incomplete oversight regimes. When the markets opened, regulatory oversight shifted from the state level to FERC. But FERC did not have real-time monitoring capabilities or an understanding of the problems that could occur in electricity markets or the legal tools to effectively prosecute deviant behavior (Wolak 2003). Because the local monitoring units in California derived their legal authority from FERC, they had no enforcement tools either. This made designing the correct incentives more crucial than in other market contexts where designers can rely on extensive analytical capabilities to record, analyze, and control transactions in real time (Stark and Pais 2020). Even when actors face problematic incentives, designers may often be able to quickly identify and constrain this behavior. This has been particularly important for electricity markets. Over time, they have become one of the most tightly surveilled types of markets in existence (García and Reitzes 2007).

That being said, oversight regimes suffer from the same basic problems as market designers. If the rules, procedures, and interfaces of the market are overly complex, the resulting market process may overwhelm the capabilities of a centralized monitoring regime. As soon as oversight fragments, it becomes difficult to observe the interaction effects between trades in different segments of the market—another version of structural secrecy (Vaughan 2008) that has been well established for the effective regulation of financial markets (MacKenzie 2011). For instance, the rise of generative AI has led to a vast increase of gaming behavior on digital labor platforms. If the range of possible behaviors in the market becomes very broad, even extensive analytical capabilities may not suffice to comb through the complexity of the marketplaces these companies manage. For this reason, there has been a "return of the state" in electricity markets despite substantial efforts to control them on a granular level (Joskow 2022). In short, while there are multiple ways in which organizations may compensate for the tension this article has identified, the article suggests that modularization is more likely to cause rather than to prevent system failures. Since only future research can truly

tell how these dynamics play out in other cases, I am now going to discuss the theoretical implications of this study.

CONCLUSIONS AND IMPLICATIONS FOR THE LITERATURE

The challenge of building and managing complex technologies has long been a focus of sociological research (Vaughan 1999; Le Coze 2015; Pettersen Gould 2021). Some research on financial crises has drawn analogies between managing complex technologies and regulating markets (Schneiberg and Bartley 2010; Min and Borch 2022). This article shows that the design of digital platform markets pushes beyond analogy. Platform markets *are* sociotechnical systems that are built and managed *by* organizations (Stark and Pais 2020). Yet, existing theories about system failures in other sociotechnical systems may not directly apply to such market design. Coordinating profit-oriented market actors poses different organizational challenges than controlling random interactions among component failures in technical systems. The article's key theoretical insight is that one of the most important strategies for avoiding system failures in safety-critical technologies—decomposing the system to empower frontline operators and designers—runs up against the inherent logic of market design and can lead to the very problems it is designed to avoid. This argument not only identifies practical limits to the logic behind modularization (Brusoni et al. 2023). It also has a basic implication for the organizational literature on system failures. It suggests that platform markets can fall prey to system failures in different ways than other organizations and that we may need new theories about strategies to avoid them. This suggests the need for research that systematically evaluates how theories about accidents in organizations hold up in the context of digital marketplaces. Particularly for the increasing number of cases where markets help run critical infrastructures for water, health care, electricity, and environmental products, this seems like an urgent project.

Even considering the extreme characteristics of the California case, the article suggests that market design is hard because designers must forgo most tools that managers in regular organizations can rely on to coordinate work. It is therefore potentially *more* difficult to plan markets that coordinate technical systems than to coordinate these systems directly. Indeed, the article hints at the possibility of a complexity threshold for the kinds of problems that digital markets can solve. As we have seen, market design may work best as centralized planning. Integrated work processes can more readily ensure globally consistent incentives than distributed work. But centralized organizations can manage complex processes only by virtue of first making them legible through methods of quantification and abstraction (Scott 1998). This makes it hard to adapt to surprising developments on the ground. It thus becomes important to understand whether modern tools

of "algorithmic management" are flexible enough to compensate for these coordination problems (Fourcade and Healy 2016; Kellogg et al. 2020). The sociology of markets and financial crises has yet to consider the limits of these new types of control from the perspective of those who deploy them. This, however, is crucial to understand the risks they present.

The article also has implications for economic sociology more generally. The literature has tended to view the order of markets as an emergent phenomenon (Fligstein and McAdam 2012; Padgett and Powell 2012). The logic of competition, cooperation, and valuation gradually evolve from the interplay between social processes surrounding the market (Beckert 2009). But the proliferation of market design in the digital economy (Viljoen et al. 2021) suggests that markets increasingly become the product of organizational planning (Stark and Pais 2020). Future research should expand on this theme in two ways.

First, it should reevaluate hypotheses about the development of markets. For instance, field theory posits that markets gradually evolve to more stability as competing firms seek to establish their "conception of control" on the institutional structure of the market (Fligstein 2001). Yet, competition in platform markets often seems to evolve the other way around. Platforms move from highly restrictive architectures with very few participating firms to more open ecosystems. Future research in the sociology of markets should thus examine how the goals of platform organizations shape and mediate the evolution of these markets. Incidentally, this involves understanding how organizational dynamics in the platform organization account for the distributional outcomes in these markets. While much research has explored undesirable dynamics in digital markets—exploitation and control of workers, racism, and various forms of aggregate inequality (Schüssler et al. 2021)—very little work has traced these developments back to decisions inside the platform organization. Implicitly, problematic developments are often attributed to a unitary, all-knowing surveillance capitalist who sets out to exploit workers. This has been recognized as a problematic fiction (Fourcade and Healy 2024). My article suggests, second, that undesirable outcomes in platform markets may reflect organizational dynamics such as incomplete coordination, structural secrecy, and temporal instability. This calls for a rapprochement between economic and organizational sociology that may be beneficial for both sides.

In sum, then, the article advocates for an economic sociology of social engineering. With the increasing digitalization of economic life, we are entering a period of unprecedented levels of social control by organizations that design and manage digital markets. Yet very little research seeks to understand under what conditions these organizational projects succeed or fail. This would require an approach that takes the plans of the designers seriously and then analyzes the organizational, political, and epistemic conditions

under which these plans can come to fruition—or fail. As the California case has shown, standard theories may not apply. Sociologists would therefore be well advised to look more deeply into the organizational processes behind this type of social engineering as more and more of our economic life migrates online.

KEY TO PRIMARY SOURCES

CPUC—Docket R.94-04-031/I.94-04-032, California Public Utility Commission Archive, San Francisco, California
CSA—California State Archives, Sacramento, CA
FERC—Federal Energy Regulatory Commission, Washington, DC.

APPENDIX A

Methodology

In analyzing the data for this study, I followed the rules of historical case studies (Abbott 2014; Yin 2017). My goal was to use the substantial record to reconstruct as closely as possible how the market design process was organized and how technical experts made decisions. From the outset, it must be noted that the documentation of these design processes is, as in all archival studies, fragmentary. Because electricity market design is a public process and thanks to 20 years of litigation, this problem is not as severe as in many other cases of market design. In particular, the California state archives have preserved a vast treasure of records that document the creation and operation of CAISO and PX. Triangulating this material with records from the other three archives also mediates the selection bias that all archival studies have to grapple with. That being said, there are still gaps in the record. Some of CAISO's and PX's business documents are withheld from the public because the material is still relevant to litigation. Some of FERC's documents have been redacted as the result of settlements with power marketers. Very little evidence remains of the meetings from the WEPEX steering committee. In the analysis, I try to address this issue by citing direct evidence for claims about organizational processes wherever possible. In some cases, I had to rely on the secondary literature for details.

To handle the vast amount of data from the archives, the analysis proceeded in several stages. I began by reconstructing the operation of the imbalance and financial markets in detail. I read operating manuals, internal reports, and design documents related to the Ricochet litigation. This revealed the structural features that incentivized Ricochet and oriented the rest of the analysis. In the next step, I needed to trace these features to the underlying design process. I began by filtering the archival material for information about the organizational structure of the design process, the size of teams,

the task distribution, their interactions, and the criteria of decision-making on the different levels. Along the way, I established chronologies of decision-making, organizational charts, notes on the work culture, and the setup of teams.

In the next stage, I analyzed how the designers framed the decisions that led to the Ricochet games and how the perception of relevant issues varied across the organization. There were three locations at which designers made relevant decisions or should have detected the negative implications of such decisions: the team working on CAISO's imbalance markets, the teams working on the auction rules for the PX, and the executive level of the design process. I went through the documents to isolate anything that pertained to decision-making in these three contexts and coded the relevant documents for "definition of task," "background of expertise," "methods of analysis," "perceived problems," "organizational context," and "communication links." Interviews helped to clarify the context of relevant design documents. I then developed subcodes that captured how designers in the different positions reflected on the rules that gave rise to the illegal arbitrage games. This progressively revealed how the organizational setup shaped how teams defined problems, what they paid attention to, and what remained hidden from sight.

Throughout, I used interviews in a purely supplementary fashion. With 20 years of distance, the risk of rationalization and hindsight bias is substantial, and I used none of the transcripts to establish facts about the design process. Instead, I used the interviews to clarify the meaning of archival documents and provide context. Twenty were with trained economists, and 15 with engineers and system operations experts. The interviews lasted, on average, between 60 and 90 minutes and were conducted over the phone or in person when feasible. They were organized around a consistent set of questions but were also tailored to respondents' areas of expertise. The selection of participants followed a mix of snowball and quota sampling based on the organizational location of their work.

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